

CRYPTOGRAPHY AND NETWORK SECURITY LAB

IV B.TECH - I SEMESTER								
Course Code	Category	Hours/Week			Credits	Maximum Marks		
A6CY08	PCC	L	T	P	C	CIE	SEE	Total
		-	-	2	1	40	60	100
Tutorial Classes: Nil		Practical Classes: 21				Total Classes: 21		

COURSE OBJECTIVES

The course will enable the students to:

1. Explain the objectives of information security.
2. Explain the importance and application of confidentiality, integrity, authentication and availability.
3. Understand various cryptographic algorithms.
4. Understand Buffer OverFlow attacks, Intrusion Detection Systems and Honeypots.
5. Understand basic cryptographic algorithms, message and web authentication and security issues.

COURSE OUTCOMES

Upon successful completion of this course, student will be able to:

1. Identify information system requirements for both of them such as client and server.
2. Understand the current legal issues towards information security.
3. Understand the process of capturing Network traffic using tools(Ethereal, Wireshark, Tcpdump)
4. Implement Cryptographic algorithms in C/C++/Java
5. Create applications in Client Server architecture.
6. Set up secure mail and web communication channels

LIST OF EXPERIMENTS

WEEK-1

1. Write a C program that contains a string (char pointer) with a value 'Hello world'. The program should XOR each character in this string with 0 and displays the result.

WEEK-2

2. Write a C program that contains a string (char pointer) with a value 'Hello world'. The program should AND or and XOR each character in this string with 127 and display the result.

WEEK-3

3. Write a Java program to perform encryption and decryption using the following algorithms
 - a. Caesar cipher
 - b. Substitution cipher
 - c. Hill Cipher

WEEK-4

4. Write a C/JAVA program to implement the DES algorithm logic.

WEEK-5

5. Write a C/JAVA program to implement the Blowfish algorithm logic.

WEEK-6

6. Write a C/JAVA program to implement the Rijndael algorithm logic.

WEEK-7

7. Write the RC4 logic in Java Using Java cryptography; encrypt the text "Hello world" using Blowfish. Create your own key using Java key tool.

WEEK-8

8. Write a Java program to implement RSA algorithm.

WEEK-9

9. Implement the Diffie-Hellman Key Exchange mechanism using HTML and JavaScript.

WEEK-10

10. Calculate the message digest of a text using the SHA-1 algorithm in JAVA.

WEEK-11

11. Calculate the message digest of a text using the MD5 algorithm in JAVA.

TEXT BOOKS

1. Cryptography and Network Security - Principles and Practice: William Stallings, Pearson Education, 6th Edition.
2. Cryptography and Network Security: Atul Kahate, McGraw Hill, 3rd Edition.

REFERENCE BOOKS

1. Cryptography and Network Security: C K Shyamala, N Harini, Dr T R Padmanabhan, Wiley India, 1st Edition.
2. Cryptography and Network Security: Forouzan Mukhopadhyay, McGraw Hill, 3rd Edition.
3. Information Security, Principles, and Practice: Mark Stamp, Wiley India.
4. Principles of Computer Security: WM. Arthur Conklin, Greg White, TMH.
5. Introduction to Network Security: Neal Krawetz, CENGAGE Learning.
6. Network Security and Cryptography: Bernard Menezes, CENGAGE Learning.